

Gavin Prescott

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<https://gavinprescott.github.io/RoboticsPortfolio/>

EDUCATION

Boston University

*Master of Science, Robotics and Autonomous Systems. College of Engineering
Dean's Scholarship. GPA: 3.41/4.0.*

Boston, MA

Expected December 2027

St. Lawrence University

Bachelor of Arts, Major in Anthropology, Minor in Biology. Cum Laude. GPA: 3.50/4.00.

Canton, NY

May 2023

EXPERIENCE

Graduate Research Assistant

Boston University RASTIC Robotics Lab

May 2026 – Present

Boston, MA

- Designed a low-cost, 4-axis laser-cut robotic arm and facilitated hands-on embedded control workshops (Arduino, Python, servos) targeting high school students from underrepresented communities.
- Maintained rapid prototyping equipment (Bambu 3D printers, custom laser cutters) and mentored students through embedded hardware-software integration (Python, ESP32) while enforcing lab safety standards.

Project Engineer Intern

Martin Electric

Feb 2025 – July 2025

Rensselaer, NY

- Diagnosed, troubleshooted, repaired, and operated a plasma CNC machine to support manufacturing and fabrication processes.
- Designed a junction box in Autodesk Fusion and generated a Bill of Materials to address space organizational requirements for the electrical field team.
- Designed a storage rack for electrical transformers in Autodesk Fusion, collaborating with the project manager and fabrication specialist to select materials and optimize the design for manufacturability and function.

Engineering Intern

The Warbird Factory

March 2024 – Feb 2025

Latham, NY

- Integrated STM microcontroller boards into medical equipment hardware, performing chip removal, soldering, and serial encoder implementation to retrofit legacy systems.
- Designed a CAD bracket tray enabling efficient integration of 3 STM boards within a constrained space.
- Developed simplified design solutions for EV charging stations, reducing part count and complexity to improve ease of manufacturing and assembly while cutting material costs by 33%.
- Performed precision machining alongside mechanics to repair and fabricate custom parts for WWII aircraft, ensuring strict adherence to technical specifications.
- Supported upgrades to vintage aircraft to comply with current FAA regulations, balancing modern safety standards with preservation of historical integrity.

PROJECTS

ARCTOS Robotic Arm

- Team lead for building a 6-axis robotic arm based on the open-source ARCTOS project, applying hands-on mechanical assembly, electronics integration, and embedded systems knowledge.
- Modifying existing open source document to apply to current industry problems in inventory management using computer vision and dynamic control protocols.

ACTIVITIES & LEADERSHIP

BU Robotics Club

Vice President

June 2025 – Present

- Scaled organization from 12 to 140+ active members in one semester, establishing organizational infrastructure to support multi-team engineering projects.
- Designed a technical screening and placement system to evaluate student skill sets and placement onto specialized hardware, software, and integration sub-teams.
- Direct project roadmaps across multiple engineering groups to meet aggressive prototyping and competition deadlines.

RELEVANT COURSEWORK

Intensive Intro to Robotics | Dynamics and Control of Mechanical Systems | Engineering Mechanics 2 (Dynamics)
CAD and Machine Components | Intro to Software Engineering | Electric Circuits

SKILLS

Electronics and Embedded Systems: Arduino, Raspberry Pi, STM boards, Soldering, Servo/Stepper Motors

Programming Languages: Python, C++, MATLAB

Software: Autodesk Fusion 360, AutoCAD, SolidWorks, OnShape, Github

Prototyping and Manufacturing: 3D printing, Laser cutting, Plasma cutting, CNC mill, Drill press, Manual lathe